

Lattice Structure

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We know that $y(n) = \sum_{k=0}^{M-1} h(k) x(n-k)$ for FIR systems

$$H(z) = \sum_{k=0}^{M-1} b_k z^{-k}$$

$$h(n) = \begin{cases} b_n & 0 \leq n \leq M-1 \\ 0 & \text{otherwise} \end{cases} \rightarrow b_0 = 1 \Rightarrow h_m(0) = 1$$

Let us define the FIR filter in the following form

$$H_m(z) = A_m(z), \quad m = 0, 1, 2, \dots, M-1$$

Then $A_m(z) = 1 + \sum_{k=1}^m \alpha_m(k) z^{-k}$ $m \geq 1$ — (1)

$$A_0(z) = 1, \quad \alpha_m(k) = h_m(k)$$

Let $\{x(n)\}$ be the input sequence to the filter $A_m(z)$ and $\{y(n)\}$ be the output

- Lattice filter implementation is widely used in adaptive filtering. Assume that we have a filter with transfer function $H(z)$. We can write,

$$H_m(z) = A_m(z), \quad m = 0, 1, 2, \dots, M - 1$$

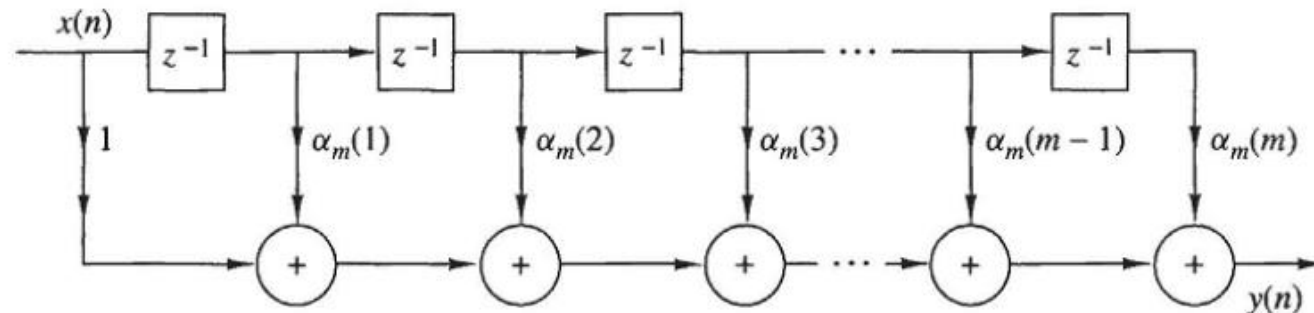
- where $A_m(z)$ is a polynomial with $A_0(z) = 1$

$$A_m(z) = 1 + \sum_{k=1}^m \alpha_m(k) z^{-k}, \quad m \geq 1$$

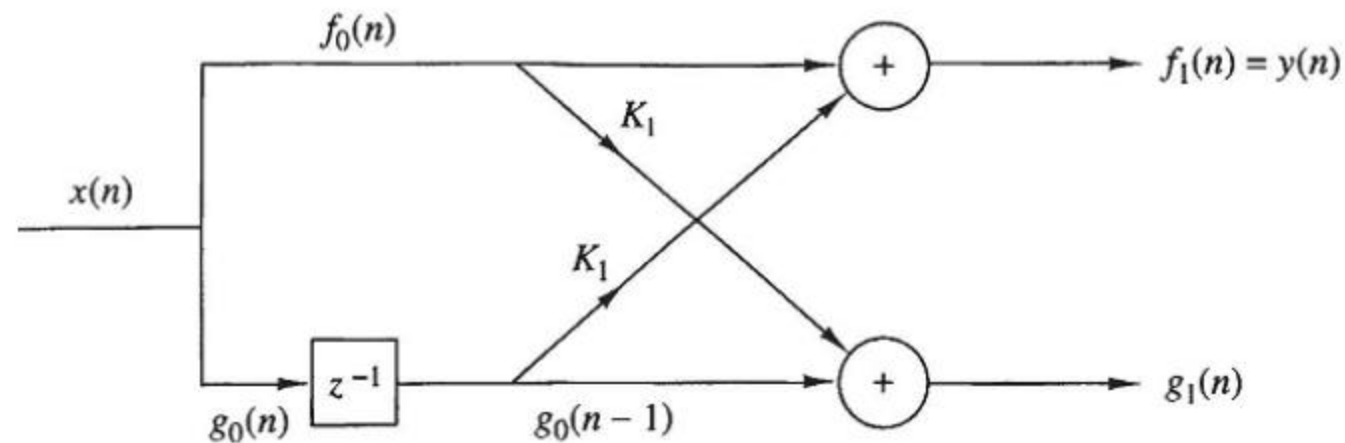
- $Y[n]$ can be written as

$$y(n) = x(n) + \sum_{k=1}^m \alpha_m(k) x(n - k)$$

- The direct form implementation can be expressed as



- Let's consider a first order FIR filter, i.e., $m=1$: $y(n] = x(n] + \alpha_1(1)x(n - 1)$
- Let the reflection coefficient $\bar{K}_1 = \alpha_1(1)$. to get:



$$\begin{aligned}
 f_0(n] &= g_0(n] = x(n] \\
 f_1(n] &= f_0(n] + K_1 g_0(n-1] = x(n] + K_1 x(n-1) \\
 g_1(n] &= K_1 f_0(n] + g_0(n-1] = K_1 x(n] + x(n-1)
 \end{aligned}$$

- Now consider $m=2$:

$$y(n) = x(n) + \alpha_2(1)x(n-1) + \alpha_2(2)x(n-2)$$

- We cascade two lattice stages:
- The output of the first stage is,

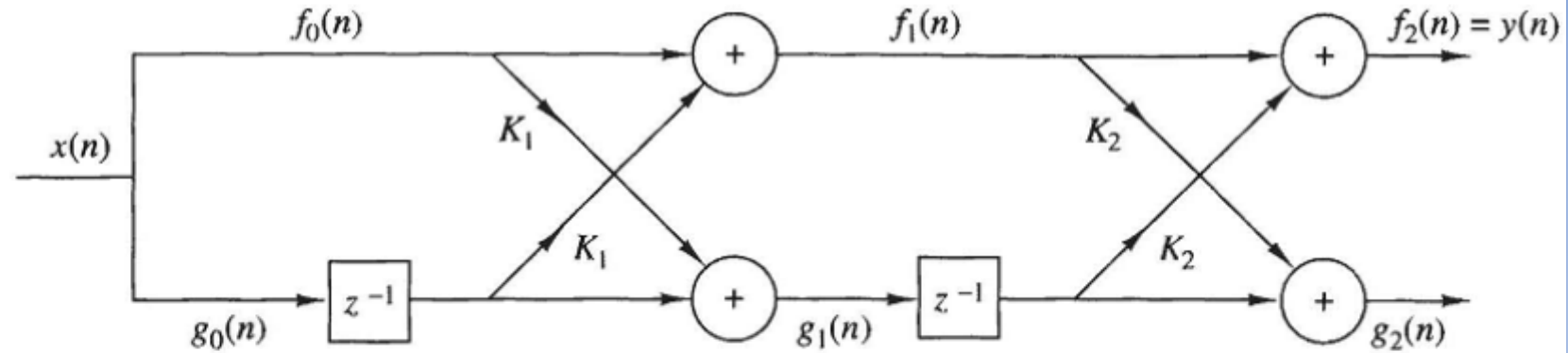
$$f_1(n) = x(n) + K_1x(n-1)$$

$$g_1(n) = K_1x(n) + x(n-1)$$

- And the output of the second stage is:

$$f_2(n) = f_1(n) + K_2g_1(n-1)$$

$$g_2(n) = K_2f_1(n) + g_1(n-1)$$



Two-stage lattice filter.

- Let's consider on $f_2[n]$:

$$\begin{aligned} f_2(n) &= x(n) + K_1 x(n-1) + K_2 [K_1 x(n-1) + x(n-2)] \\ &= x(n) + K_1(1 + K_2)x(n-1) + K_2 x(n-2) \end{aligned}$$

- $f_2[n]$ will be $y[n]$ if:

$$\alpha_2(2) = K_2, \quad \alpha_2(1) = K_1(1 + K_2)$$

- or, equivalently if:

$$K_2 = \alpha_2(2), \quad K_1 = \frac{\alpha_2(1)}{1 + \alpha_2(2)}$$

- In general:

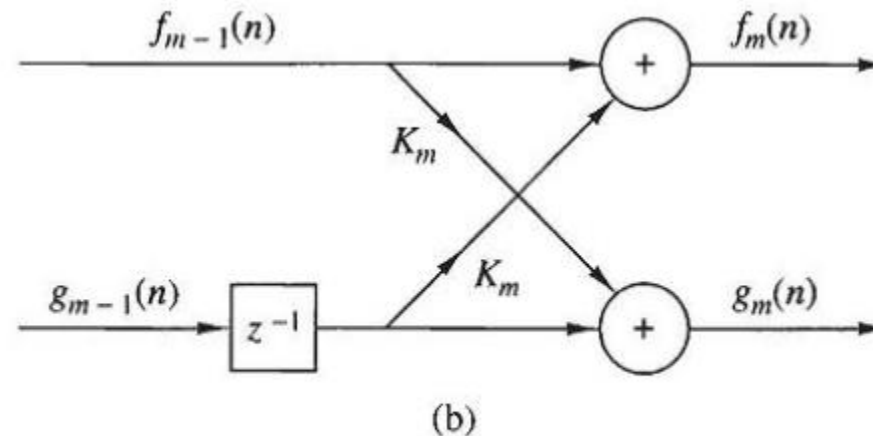
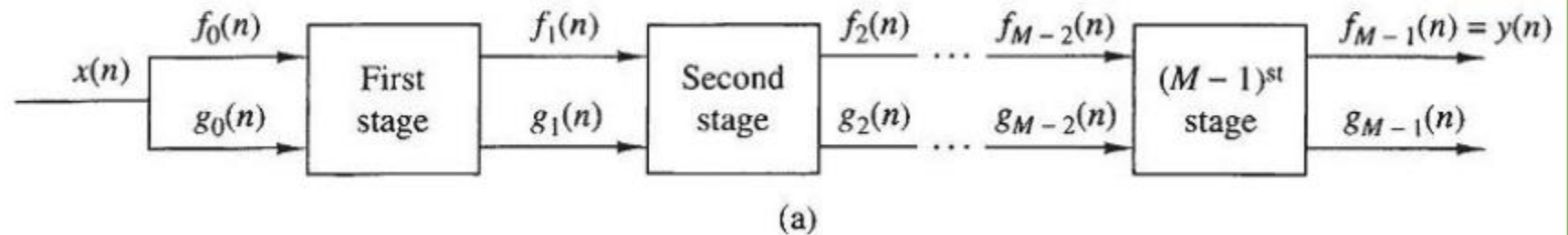
$$f_0(n) = g_0(n) = x(n)$$

$$f_m(n) = f_{m-1}(n) + K_m g_{m-1}(n-1), \quad m = 1, 2, \dots, M-1$$

$$g_m(n) = K_m f_{m-1}(n) + g_{m-1}(n-1), \quad m = 1, 2, \dots, M-1$$

- Then:

$$y(n) = f_{M-1}(n)$$



(M-1)-stage lattice filter.

Conversion of FIR taps to Lattice coefficients

$$\alpha_{m-1}(k) = \frac{\alpha_m(k) - \alpha_m(m) \alpha_m(m-k)}{1 - \alpha_m^2(m)}$$

Determine the lattice coefficients corresponding to the FIR filter with system function

$$Y(n) = x(n) + \frac{1}{3}x(n-1) + \frac{1}{2}x(n-2) + \frac{1}{3}x(n-3)$$

$$\begin{aligned}\alpha_3(0) &= 1 \\ \alpha_3(1) &= \frac{1}{3} \\ \alpha_3(2) &= \frac{1}{2} \\ \alpha_3(3) &= \frac{1}{3}\end{aligned}$$

$$\begin{aligned}k_m &= \alpha_m(m) \\ k_1 &= \alpha_1(1) \\ k_2 &= \alpha_2(2) \\ k_3 &= \alpha_3(3)\end{aligned}$$

$$\alpha_{m-1}(k) = \frac{\alpha_m(k) - \alpha_m(m)\alpha_m(m-k)}{1 - \alpha_m^2(m)}$$

$$k_3 = \alpha_3(3) = \frac{1}{3}$$

$$\begin{aligned}k_2 &= \alpha_2(2), \quad m=3, k=2 \\ \alpha_2(2) &= \frac{\alpha_3(2) - \alpha_3(3)\alpha_3(1)}{1 - \alpha_3^2(3)}\end{aligned}$$

$$k_2 = \frac{2}{16}$$

$$\alpha_2(1) = m=3, k=1$$

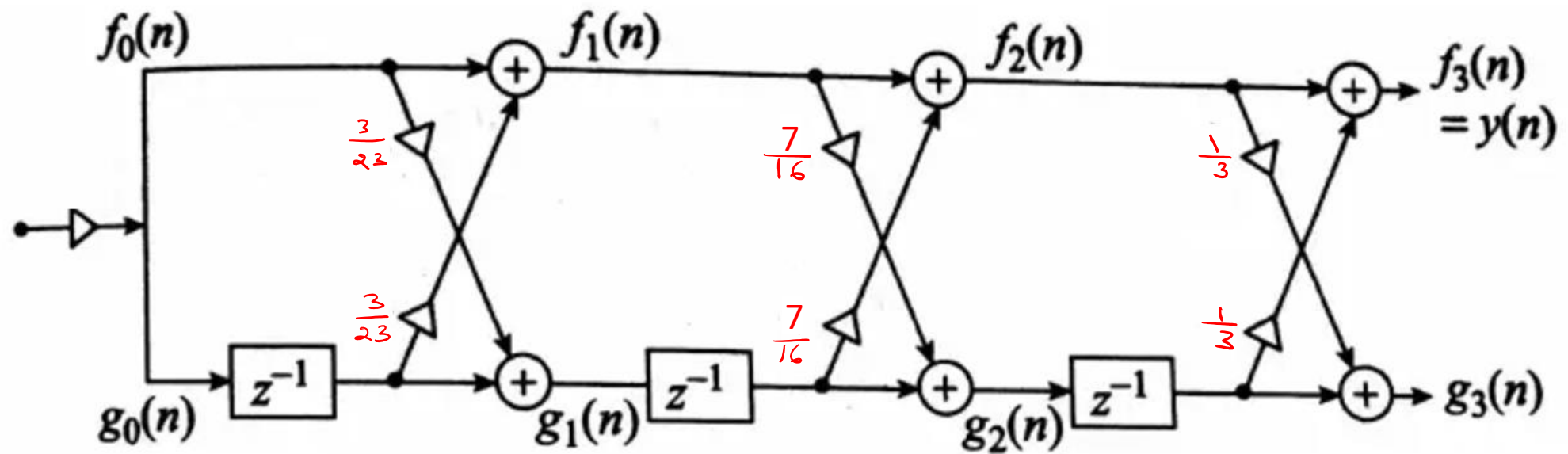
$$\alpha_2(1) = \frac{\alpha_3(1) - \alpha_3(3)\alpha_3(2)}{1 - \alpha_3^2(3)}$$

$$k_2(1) = \frac{3}{16}$$

$$k_1 = \alpha_1(1), \quad m=2, k=1$$

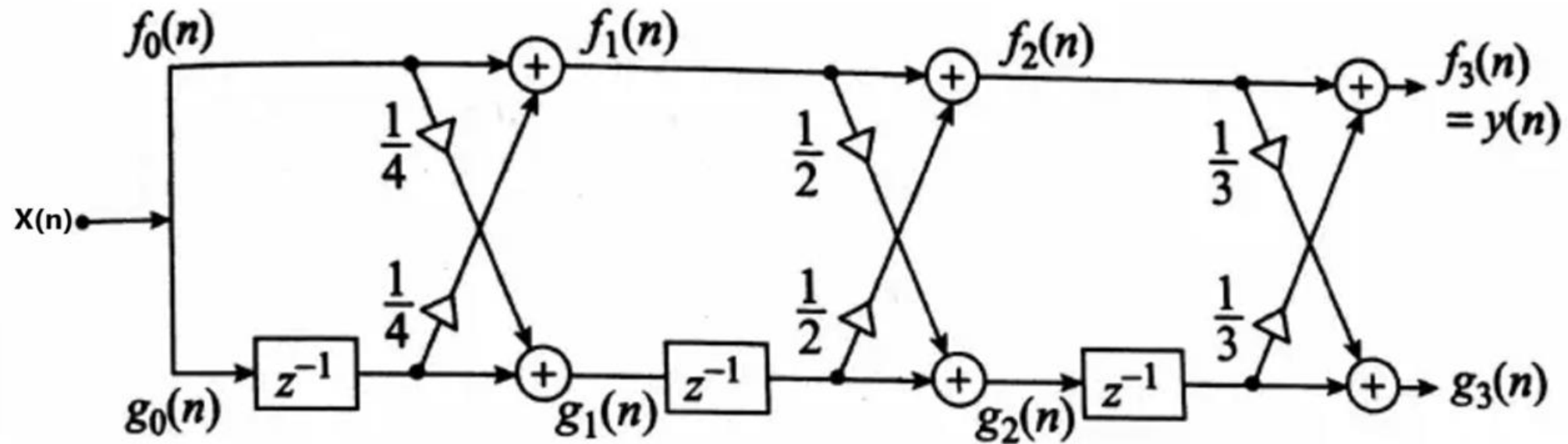
$$\alpha_1(1) = \frac{\alpha_2(1) - \alpha_2(2)\alpha_2(1)}{1 - \alpha_2^2(2)}$$

$$= \frac{3}{23}$$



Determine the lattice coefficients corresponding to the FIR filter with system function

$$H(z) = A_3(z) = 1 + \frac{13}{24}z^{-1} + \frac{5}{8}z^{-2} + \frac{1}{3}z^{-3}$$



Consider a three – stage FIR lattice structure having the coefficients
 $k_1=0.65$, $k_2=0.5$ & $k_3=0.9$
Find its impulse response and direct form structure.

$$\alpha_m(0) = 1 \quad \text{—————} \quad \textcircled{\text{I}}$$

$$\alpha_m(m) = k_m \quad \text{—————} \quad \textcircled{\text{II}}$$

$$\alpha_m(k) = \alpha_{m-1}(k) + k_m \alpha_{m-1}(m-k) \quad \text{—————} \quad \textcircled{\text{III}}$$

$$\boxed{m=3}$$

$$\boxed{\alpha_m(0) = 1}$$

$$\boxed{\alpha_1(1) = 0.65}$$

$$\boxed{\alpha_2(2) = 0.5}$$

$$\boxed{\alpha_3(3) = 0.9}$$

To realize direct form structure we need to find

$$\underline{\alpha_3(1)}$$

$$\underline{\alpha_3(2)}$$

$$\underline{\alpha_3(3)}$$

$$\alpha_m(k) = \alpha_{m-1}(k) + k_m \alpha_{m-1}(m-k) \quad \text{--- (III)}$$

put $m=3, k=1$ in Eqn (III),

$$\alpha_3(1) = \alpha_2(1) + k_3 \times \alpha_2(2) \quad \text{--- (A)}$$

Now calculate $\alpha_2(1)$

put $m=2, k=1$ in Eqn (III)

$$\alpha_2(1) = \alpha_1(1) + k_2 \times \alpha_1(1) = 0.65 + 0.5 \times 0.65 = 0.975$$

Substitute $\alpha_2(1)$ in Eqn (A)

$$\therefore \alpha_3(1) = 0.975 + 0.9 \times 0.5 = 1.425$$

Now substitute $m=3, k=2$ in Eqn (III)

$$\alpha_3(2) = \alpha_2(2) + k_3 \times \alpha_2(1) = 0.5 + 0.9 \times 0.975 = 1.3775$$

$$\alpha_3(1) = 1.425$$

$$\alpha_3(2) = 1.3775$$

$$\alpha_3(3) = 0.9$$

$\therefore$ w.k.t

$$H(z) = 1 + \sum_{k=1}^m \alpha_m(k) z^{-k}$$

$$H(z) = 1 + \sum_{k=1}^3 \alpha_m(k) z^{-k}$$

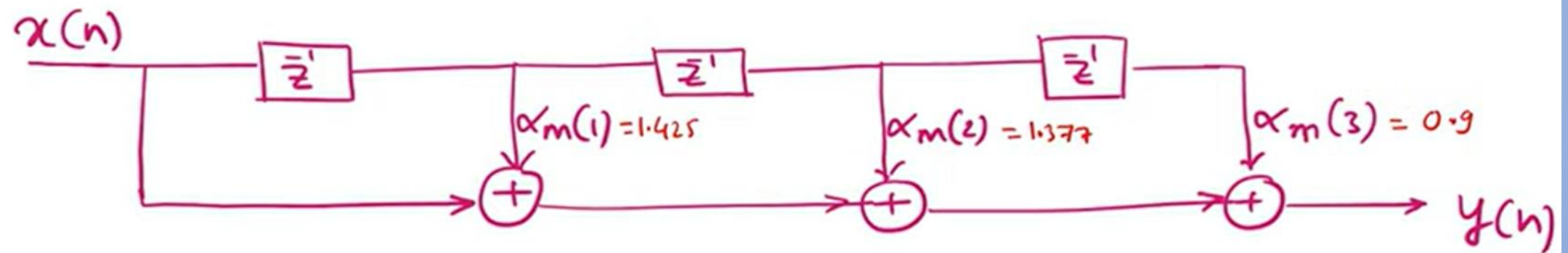
$$H(z) = 1 + \alpha_3(1) z^{-1} + \alpha_3(2) z^{-2} + \alpha_3(3) z^{-3}$$

$$\therefore H(z) = 1 + 1.425 z^{-1} + 1.3775 z^{-2} + 0.9 z^{-3}$$

$$\frac{Y(z)}{X(z)} = 1 + 1.425 z^{-1} + 1.3775 z^{-2} + 0.9 z^{-3}$$

$$\boxed{y(n) = x(n) + 1.425 x(n-1) + 1.3775 x(n-2) + 0.9 x(n-3)}$$

Direct form structure is shown below

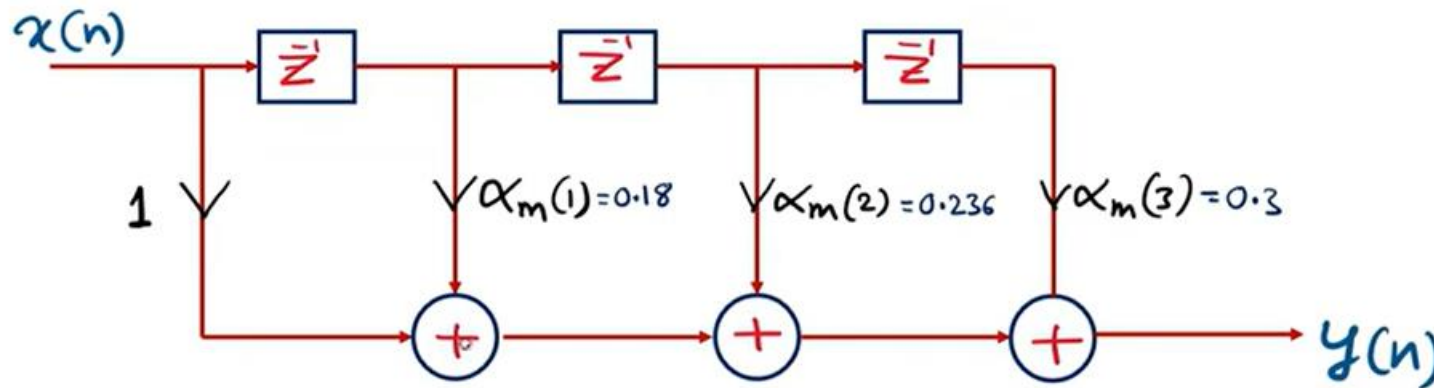


Let the coefficients of a three stage FIR lattice structure be

$$K_1=0.1, K_2=0.2 \text{ and } K_3=0.3.$$

Find the coefficients of the direct form 1 FIR filter and draw its block diagram

Direct form structure



*Thank
you*

